Assignment 2 Instructions (Due Week 8)

Listen

Assignment 2: Online Cruise Booking System (Spring Boot Web App)

Due Date: March 10 at 11:59 pm

Assessment Information

Total Mark: 60 marks - 15% of the overall grade

Purpose:

Develop a Spring Boot-based web application for a **Luxury Cruise Web App Development**, implementing **CRUD functionality** using **JPA and Hibernate**. The application should connect to a **MySQL** database and provide a user-friendly interface for customers to book, view, modify, and cancel flight reservations.

By finishing this assignment, you will practice to:

- Design, code and test a Spring Boot MVC app with Thymeleaf template and Spring Data JPA.
- Design and Implement MySQL database to connect with Spring Boot JPA.

Read the lecture slides week 3 – 6 and lab exercises. This material provides the necessary information you need to complete the exercises.

Instructions and Rules

- This lab should be completed by every student **individually or in pairs.**
- You will have to **demonstrate your solution in a scheduled lab session** and submitting the project through the assignment drop box on Luminate. You must name your Eclipse project according to the following rule:
- Your Name_COMP303_AssignementNumber
    - Example: **John_COMP303_Assignment2**

Functional Requirements:

1. User Login and Registration

The Home (Index) Page should include:

- A **"Sign In"** option for existing users.
- A **"Create Account"** link for new customers to register.

2. Cruise Reservation

Upon successful login, customers will be redirected to the **Cruise Reservation Page**, where they can:

- Select **cruise departure and return dates**.
- Choose **departure port and destination**.
- Select **cabin type** (e.g., Interior, Ocean View, Balcony, Suite).
- Enter the number of passengers (adults and children).
- View **cruise price**, **ship image/logo**, itinerary highlights, and onboard amenities (preferred).

3. Payment Integration

After selecting cruise details, customers proceed to checkout.

- Provide options for **credit/debit card, PayPal, Apple Pay, or Google Pay**.
- **Note:** Payment details should **NOT** be stored in the database.

4. Payment Confirmation

- Display a **booking confirmation message** (on-page, alert, or dialog box) upon successful payment.
- Confirmation should include **booking reference number**, cruise name, and departure date.

5. Modify Reservation

Customers should be able to:

- Edit their reservation details, such as:
    - Cruise dates
    - Cabin type
    - Passenger count

6. Cancel Reservation

- Allow customers to cancel their cruise reservation **up to 10 days before departure**.
- Implement **date validation logic** to enforce this cancellation policy.

7. Customer Profile Management

Customers can access their profile page to:

- View and update personal information such as:
    - Address
    - Phone number
    - Email

- Password

8. Server-Side Validation

- Server-side validation should be implemented using appropriate **annotations in the entity classes**.
- Display validation errors using **error handling tags in Thymeleaf templates**.

9. Web Interface Design

Design a clean, accessible, and user-friendly interface using:

- **Thymeleaf HTML files**
- **CSS or Bootstrap** for styling
- Include **luxury cruise imagery, ship visuals, and branding** for a premium, professional appearance.

10. Database Requirements (MySQL)

Create a relational database named **CruiseReservation** with at least the following main tables:

- **Customer –** customer and traveler details
- **Cruise –** cruise itinerary, ship, ports, dates, and pricing
- **Booking –** reservation details linking passengers and cruises

Customer	Cruise	Booking
customer_id username or email password firstname lastname address city postalCode	cruise_id cruise_name departure_time arrival_time origin destination price	booking_id customer_id (FK) cruise_id (FK) booking_date departure_date no_of_passengers total_price status

Assessment Rubrics

Description	Points

Functionalities: Developing a Spring Boot MVC to satisfy all the requirements (1-7)	30
MySQL database design and use of JPA/CRUD repositories (10)	12
UI friendliness (proper layout, colors, fonts, use of CSS) and page navigation (9)	6
Input form validations including Entity class annotations and Thymeleaf validations and Code stands, use of variables, comments, and naming conventions. (8)	6
Lab demo	6
Total	**60**

Submission Rules

Each file submitted in the solution should have student name, student number and submission date in the top of the code file. (Code standard)

Comments may be necessary inside the functions or predicates but that the names of the functions and predicates and the comments you include to explain each are especially important. (Code standard)

Each students must hand in the assignment to Luminate drop box.

Late Submission
There is a 20% off per day late penalty which means if a student submits an assignment one day late and does perfect work, the most they can obtain is 80/100.

For example, if a student submits one day late and obtains 8/10 as an assignment then the professor will need to deduct 20% from this grade (8 x .20 = 1.6, 8-1.6 = 6.4/10 - record 6.4). If a student submits two days late then 40% will be deducted, for example, assume the student obtains 9/10 (9 x .40 = 3.6, 9-3.6 = 5.4/10). By day five the student receives zero.

Academic Honesty (Plagiarism and Cheating)
All students must follow the academic honesty policies regarding Plagiarism and cheating on assignments, Quizzes or Tests. Centennial college's Academic Policy will be strictly enforced. To support academic honesty at Centennial College, all academic work submitted by students may be reviewed for authenticity and originality, with utilizing software tools.